

Paper 1 - Set 02 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	C	Simultaneous equations and inequalities
2	A	Angles, polygons, congruence and vectors
3	F	GCSE algebra
4	C	Exponentials and logarithms
5	A	Functions and transformations
6	B	Trigonometric functions and equations
7	F	Differentiation
8	A	Quadratic functions
9	C	Triangles, radians and sectors
10	E	Probability
11	F	Number, units and estimation
12	D	Functions and transformations
13	D	Trigonometric functions and equations
14	C	Simultaneous equations and inequalities
15	F	Sequences and series
16	E	Polynomials, factor and remainder theorems
17	D	Integration
18	A	Exponentials and logarithms
19	E	Quadratic functions
20	A	Graphs of functions

Worked solutions

Question 1 SIM-09 · Simultaneous equations and inequalities · spec MM1.4

The simultaneous equations $y = x^2 + 2x + c$ and $y = 4x + 1$ have exactly one solution. Find the value of c .

Answer: C

$$x^2 - 2x + (c - 1) = 0 \text{ has one solution when } 4 - 4(c - 1) = 0, \text{ so } c = 2.$$

Question 2 GEO-08 · Angles, polygons, congruence and vectors · spec M5.4

Which of the following sets of information is NOT sufficient to prove that two triangles are congruent?

Answer: A

SSA is not a congruence condition (it is the ambiguous case of the sine rule); the other four are the standard criteria.

Question 3 ALG-12 · GCSE algebra · spec M4.19

A sequence has first term 2 and each subsequent term is obtained by squaring the previous term and subtracting 2. What is the 50th term?

Answer: F

$$2^2 - 2 = 2: \text{ every term is } 2.$$

Question 4 LOG-11 · Exponentials and logarithms · spec MM5.2

Solve $\log_2 x + \log_4 x = 6$. (You may use $\log_4 x = \frac{1}{2}\log_2 x$.)

Answer: C

$$\frac{3}{2}\log_2 x = 6 \Rightarrow \log_2 x = 4 \Rightarrow x = 16.$$

Question 5 FUN-13 · Functions and transformations · spec MM8.4

The graph of $y = a(x + b)^2 + c$ has vertex $(-2, 5)$ and passes through $(0, -3)$. Find a .

Answer: A

$$b = 2, c = 5. \text{ At } x = 0: 4a + 5 = -3, \text{ so } a = -2.$$

Question 6 TRG-12 · Trigonometric functions and equations · spec MM4.6

Find the number of solutions of $\tan^2 x = 3$ for $0 \leq x < 2\pi$.

Answer: B

$$\tan x = \pm\sqrt{3}: x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}. \text{ Four solutions.}$$

Question 7 DIF-03 · Differentiation · spec MM6.3

The tangent to $y = x^2 - 4x + 1$ at the point where $x = 3$ meets the y-axis at

Answer: F

$$y' = 2x - 4 = 2 \text{ at } x = 3; \text{ the point is } (3, -2). \text{ Tangent: } y + 2 = 2(x - 3), \text{ so at } x = 0, y = -8.$$

Question 8 QUA-21 · Quadratic functions · spec MM1.3

The parabola $y = x^2 - 4x + 3$ is reflected in the x -axis and then translated 2 units in the positive y -direction. What is the maximum value of the resulting function?

Answer: A

$x^2 - 4x + 3 = (x - 2)^2 - 1$, minimum -1 . Reflecting gives maximum 1; translating up by 2 gives maximum 3.

Question 9 TRI-13 · Triangles, radians and sectors · spec MM4.2

The minute hand of a clock is 12 cm long. How far does its tip travel in 25 minutes?

Answer: C

25 minutes is $\frac{25}{60} = \frac{5}{12}$ of a revolution: $\frac{5}{12} \times 2\pi \times 12 = 10\pi$ cm.

Question 10 PRO-11 · Probability · spec M7.6

Two cards are drawn at random from a standard pack of 52 without replacement. Which of the following expressions is the probability that both are aces?

Answer: E

Without replacement the second draw has 3 aces among 51 cards.

Question 11 NUM-15 · Number, units and estimation · spec M2.5

How many three-digit positive integers have digits that are all different and all odd?

Answer: F

Odd digits 1, 3, 5, 7, 9: $5 \times 4 \times 3 = 60$.

Question 12 FUN-11 · Functions and transformations · spec MM8.2

The graph of $y = f(x)$ is transformed by the sequence: reflect in the x -axis, then translate 2 units in the positive y -direction, then reflect in the y -axis. The resulting graph has equation

Answer: D

Reflect in x -axis: $y = -f(x)$. Translate up 2: $y = -f(x) + 2$. Reflect in y -axis (replace x by $-x$): $y = 2 - f(-x)$.

Question 13 TRG-13 · Trigonometric functions and equations · spec MM4.5

For all θ where the expressions are defined, $(\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2$ equals

Answer: D

Expanding: $(1 + 2\sin \theta \cos \theta) + (1 - 2\sin \theta \cos \theta) = 2$.

Question 14 SIM-12 · Simultaneous equations and inequalities · spec MM1.4

The curves $y = x^2 - 4$ and $y = -x^2 + 2x + k$ intersect at exactly two points. Find the complete set of values of k .

Answer: C

$2x^2 - 2x - 4 - k = 0$: discriminant $4 + 8(4 + k) > 0 \Rightarrow 36 + 8k > 0 \Rightarrow k > -\frac{9}{2}$.

Question 15 SEQ-08 · Sequences and series · spec MM2.3

The second term of a geometric sequence is 6 and the fifth term is 48. Find the sum of the first 6 terms.

Answer: F

$$ar = 6, ar^4 = 48, \text{ so } r^3 = 8, r = 2, a = 3. S_6 = 3 \cdot \frac{2^6 - 1}{2 - 1} = 3 \times 63 = 189.$$

Question 16 POL-13 · Polynomials, factor and remainder theorems · spec MM1.6

Which of the following is a factor of $x^4 - 5x^2 + 4$?

Answer: E

$$x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4) = (x - 1)(x + 1)(x - 2)(x + 2).$$

Question 17 INT-15 · Integration · spec MM7.1

The area between the curve $y = x^2$ and the line $y = k$ (where $k > 0$) is $\frac{32}{3}$. Find k .

Answer: D

$$\text{Area} = \int_{-\sqrt{k}}^{\sqrt{k}} (k - x^2) dx = 2 \left(k^{3/2} - \frac{k^{3/2}}{3} \right) = \frac{4}{3} k^{3/2} = \frac{32}{3} \Rightarrow k^{3/2} = 8 \Rightarrow k = 4.$$

Question 18 LOG-14 · Exponentials and logarithms · spec MM5.3

The number of integers n satisfying $2^n < 1000 < 2^{n+3}$ is

Answer: A

$$2^9 = 512 < 1000 < 1024 = 2^{10}. \text{ Need } n \leq 9 \text{ and } n + 3 \geq 10, \text{ so } n = 7, 8, 9; \text{ three integers.}$$

Question 19 QUA-09 · Quadratic functions · spec MM1.3

Find the sum of all real values of x satisfying $(x^2 - 3x)^2 - 2(x^2 - 3x) - 8 = 0$.

Answer: E

$$\text{Let } u = x^2 - 3x: (u - 4)(u + 2) = 0. x^2 - 3x - 4 = 0 \text{ gives } x = 4, -1 \text{ (sum 3). } x^2 - 3x + 2 = 0 \text{ gives } x = 1, 2 \text{ (sum 3). Total 6.}$$

Question 20 GRA-11 · Graphs of functions · spec MM8.7

The curves $y = 2^x$ and $y = x^2$ intersect at how many points?

Answer: A

$$\text{They meet at } x = 2 \text{ and } x = 4 \text{ (} 4 = 4, 16 = 16 \text{), and once for negative } x \text{ (at } x = 0: 1 > 0; \text{ at } x = -1: \frac{1}{2} < 1 \text{). For } x > 4, 2^x \text{ grows faster. Three intersections.}$$